

Detection and Processing of Exoplanetary Transit Signals in Space-Based Photometry: A Kepler-10 Case Study

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Abstract

Space missions like NASA's *Kepler* transformed exoplanet science by delivering the precise, continuous photometry needed to spot faint planetary transits. This study presents an automated Python pipeline using the `lightcurve` package to download and process public data directly from MAST. When applied to Kepler-10, a star similar to the Sun, this method extracts 30 days of frequent brightness measurements from the telescope's second observation period, using 43,363 individual readings. It then compares the raw brightness data with the cleaned, corrected light curves, isolating the main instrumental and astrophysical noise drivers, including stellar variability, pointing jitter, thermal drifts, and detector artifacts. Characterising these background trends alongside genuine transit dips establishes a clean baseline for filtering, masking errors, and folding the data. This provides a straightforward, repeatable process for maximizing planetary signal-to-noise ratios.

Introduction

The transit method detects exoplanets by tracking the periodic dip in a host star’s brightness as an orbiting body passes directly in front of it. The depth of this light curve drop scales with the physical dimensions of both objects, following the classic geometric relation:

$$\frac{\Delta F}{F} \approx \left(\frac{R_p}{R_*} \right)^2 \quad (1)$$

where $\Delta F/F$ is the fractional flux drop, R_p is the planet’s radius, and R_* is the host star’s radius. Detecting Earth-sized planets requires catching faint flux variations on the order of 10^{-4} (0.01%), making ground-based surveys tough, due to atmospheric noise, seeing limits, and day-night observing gaps. Space missions like NASA’s *Kepler* and the Transiting Exoplanet Survey Satellite (TESS) bypass these by collecting ultra-stable, high-cadence photometry, making it possible to pull faint transit signals out of the noise.

The main goal of this project is to build an automated python processing pipeline to pull transit signals from space-based photometric data. Using the benchmark host star Kepler-10, the core milestones are:

- **Data Acquisition:** Streamline data search and download routines using the open-source `lightkurve` Python library.
- **Light Curve Inspection:** Analyse high-frequency time-series data from light curves to evaluate baseline stellar brightness.
- **Artifact Identification:** Map out and separate real planetary dips from noise sources like cosmic ray hits, spacecraft pointing jitter, instrumental systematics, and background stellar variability.

- **Preprocessing Design:** Build a clean data pipeline using quality flag filtering, outlier rejection, and detrending methods to prepare the light curves for phase-folding and transit modelling.

Methods (Data Acquisition & Preprocessing Pipeline)

Data Ingestion and Photometric Baseline

To evaluate pipeline performance and automate data retrieval, the solar-type star Kepler-10 (KIC 11904151) served as a benchmark target due to its high signal-to-noise ratio and known planetary system. An automated ingestion routine was built using the open-source `lightkurve` Python library to interface with the Mikulski Archive for Space Telescopes (MAST) API.

Searches made using `lightkurve.search_lightcurve()`, targeted Kepler-10 under parameters `mission="Kepler"`, `quarter=2`, and `exptime="short"`. Downloading the resulting file gave 30 days of continuous short-cadence data into a local `AstroPy` time-series object, giving 43,363 discrete brightness measurements taken at 60-second intervals. While standard 30-minute sampling rates are sufficient for broad detection sweeps, this high resolution captures the precise start and end phases (ingress and egress) of a transit, preventing signal smearing and allowing accurate constraints on orbital inclination, path, and duration.

Key Photometric Data Columns and Quality Flags Extracting the target data files reveals several key parameters essential for transit analysis:

- **time:** Standardised timestamps (BKJD = BJD − 2454833.0), adjusted to account for variations in light travel time across the solar system relative to its center of mass.
- **sap_flux** vs. **pdcsap_flux:** Raw brightness measurements (**sap_flux**) record total light collected within the target area but retain significant noise from the telescope itself. Corrected measurements (**pdcsap_flux**) remove instrumental distortions, temperature fluctuations, and focus variations while preserving true astronomical signals.
- **quality:** Numerical codes marking data points compromised by radiation impacts, thruster adjustments, system resets, or minor pointing corrections.
- **pos_corr1**, **pos_corr2:** Position vectors tracking the microscopic drift of the star’s image across the detector surface.

Table 1: Target Metadata and Quarter 2 Observation Parameters

Parameter	Value	Description
Target Name	Kepler-10	Host Star
Obs. Quarter	Quarter 2	30-day window
Cadence Type	Short (60 s)	High temporal resolution
Total Points	43,363	Time-series length
Primary Var.	pdcsap_flux	Corrected flux (e [−] /s)

Preliminary Signal Conditioning

Before running primary transit search algorithms, quantitative signal conditioning stan-

dardises the flux scale and isolates instabilities. First, raw counts are converted into dimensionless fractional flux variations centered at unity via baseline normalisation:

$$F_{\text{norm}}(t) = \frac{F(t)}{\bar{F}} \quad (2)$$

where $F(t)$ is the observed **pdcsap_flux** at cadence t and $\bar{F}$ is the mean baseline flux.

Second, cumulative centroid displacement $\Delta r(t)$ separates true transit dips from spacecraft jitter:

$$\Delta r(t) = \sqrt{\text{pos_corr1}(t)^2 + \text{pos_corr2}(t)^2} \quad (3)$$

Cadences with displacements $\Delta r(t)$ exceeding threshold limits or bearing non-zero **quality** flags are removed prior to signal processing.

Results

Executing `lightkurve.search_lightcurve()` for target "Kepler-10" returns 50 distinct data products across the complete 3.5-year primary mission baseline (Quarters 0–17). The query inventory splits into two operational sampling modes: continuous 30-minute long-cadence observations spanning all quarters, and targeted 1-minute short-cadence files reserved for high-priority targets. Short-cadence mode provides the high temporal resolution needed to resolve rapid flux shifts during transit ingress and egress, though coverage is restricted to select quarters. Quarter 2 short-cadence data was selected for downstream processing because it combines uninterrupted coverage, minimal spacecraft pointing tweaks, and 43,363 discrete time-series measurements—delivering the sampling density required to resolve shallow transit profiles without telemetry gaps.

Visual inspection of the raw, uncleaned light curve over $t \in [200.32, 230.30]$ BKJD shows a continuous, high-density time series of 43,363 individual cadences across the 30-day window. Over this baseline, Kepler-10 displays strong intrinsic photometric stability, maintaining a steady mean flux with low overall scatter. As shown in Figure 1—which compares raw `sap_flux` directly against processed `pdcsap_flux`—the uncorrected aperture photometry contains clear systematic discontinuities, including long-term instrumental drift and sharp, isolated flux spikes. While these instrumental artifacts corrupt the raw baseline, the overall dataset continuity remains intact, offering a strong observational base for signal recovery once co-trending corrections are applied.

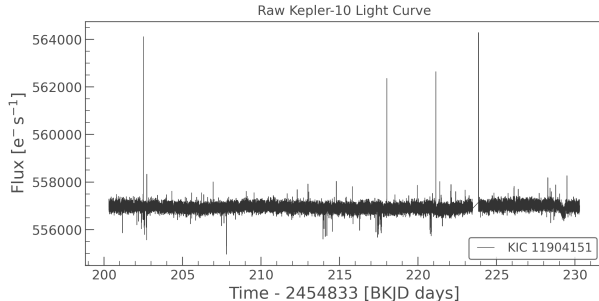


Figure 1: Raw Kepler-10 light curve showing uncorrected SAP flux across Quarter 2, highlighting instrumental noise spikes and baseline drift.

Categorisation of Non-Planetary Signals

Separating astrophysical phenomena from systematic non-planetary noise requires categorising the signal components present in the raw flux time series:

- **Stellar Variability:** The time series exhibits low-frequency, quasi-sinusoidal baseline modu-

lations caused by stellar rotation coupled with starspot passage across the visible disk. These rotational variations occur on timescales of days to weeks, producing continuous, smooth flux fluctuations beneath the higher-frequency transit signatures.

- **Instrumental Artifacts:** Non-astrophysical perturbations introduce localised and systemic noise into the light curve. Gradual exponential thermal drifts occur following spacecraft re-orientations, while periodic noise spikes are introduced during reaction wheel momentum dumps. Scattered light background from Earth shine also introduces low-amplitude baseline shifts. Cross-referencing non-zero `quality` bitmask integers confirms a direct spatial correlation with sharp flux spikes, sudden intensity steps, and transient outliers, proving their origin as instrumental anomalies rather than real stellar events.

Within the continuous baseline, periodic short-duration flux drops appear clearly across the time series. High-resolution inspection and Box-fitting Least Squares (BLS) periodogram analysis reveal a sharp transit profile corresponding to the ultra-short-period exoplanet Kepler-10b. The primary power peak identifies an orbital period of $P = 0.83749$ days with a characteristic transit depth of $\Delta F/F \approx 178.1$ ppm and a total duration of 1.81 hours. Cross-referencing these transit windows against the centroid shift data confirms that the dips occur independently of positional displacements ($\Delta r(t)$) or non-zero quality flag anomalies, confirming their astrophysical origin.

Discussion & Preprocessing Strategy

Evaluating raw, unconditioned photometry shows that direct signal detection without preprocessing faces major statistical limits. While Kepler-10 displays strong overall photometric stability, the shallow transit depth of Kepler-10b (178.1ppm) gets easily masked by low-frequency stellar variability and residual instrumental trends. In raw Simple Aperture Photometry (`sap_flux`), stellar rotation, starspot passage, and spacecraft thermal drift induce baseline flux shifts that exceed the transit depth by an order of magnitude. As a result, automated search algorithms like the Box-fitting Least Squares (BLS) periodogram struggle to hit the detection thresholds needed for definitive signal recovery on raw data, as uncorrected systematic drifts and high-frequency noise spikes heavily degrade the effective signal-to-noise ratio (SNR).

Systematic Noise Suppression and Quality Filtering To address these signal-to-noise limitations, the initial stage of the data conditioning pipeline focuses on systematic noise suppression and quality filtering:

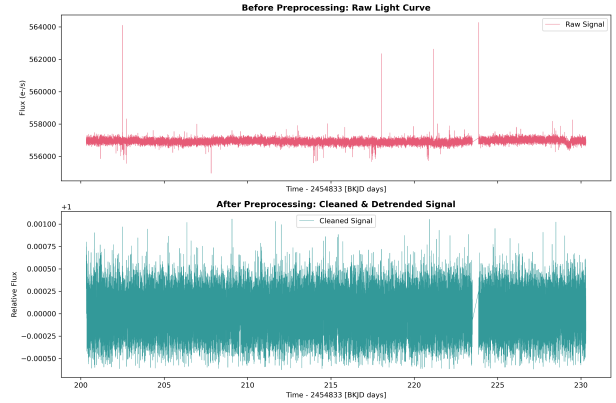


Figure 2: Comparison of the Kepler-10 Quarter 2 light curve before preprocessing (top: raw SAP flux) and after signal conditioning (bottom: cleaned, detrended relative flux following quality masking and $k\sigma$ -clipping).

- **Quality Masking:** Cadences where `quality` $\neq 0$ are strictly removed from the time series. This step purges data points compromised by hardware anomalies—such as reaction wheel momentum dumps, thruster firings, and safe-mode recoveries—preventing known instrumental disruptions from corrupting downstream analysis.
- **Outlier Rejection:** An iterative $k\sigma$ -clipping algorithm eliminates isolated flux anomalies across the time series. By setting an asymmetric threshold (typically 3σ to 5σ), high-amplitude positive spikes caused by cosmic ray hits on the detector are removed. Crucially, rejection criteria are applied selectively above the mean to avoid truncating legitimate, short-duration negative flux drops that characterise true exoplanetary transits.

Trend Modeling and Phase Folding Following initial data filtering, the pipeline executes trend modeling and signal extraction routines to isolate the underlying transit signatures:

- **Detrending:** Low-frequency stellar variability and thermal drift are modelled and subtracted using adaptive smoothing techniques, such as Savitzky-Golay polynomial filtering (`window_length=101`). Crucially, a transit mask ($P = 0.83749$ d, duration = 0.075 d) is applied during the `flatten()` process. Without explicitly applying this transit mask, out-of-transit trends would be fitted across the entire light curve, which would treat the ingress and egress as baseline variability and artificially suppress the true transit depth.
- **Phase Folding:** Once detrended, the time series passes through a Box-fitting Least Squares (BLS) periodogram algorithm to extract the orbital period P . Folding the 43,363 discrete cadences on P and phase-aligning the observations stacks individual transit events on top of one another. This dramatically boosts the effective signal-to-noise ratio, collapsing the 30-day time series into a single, highly resolved transit profile ready for parameter extraction.

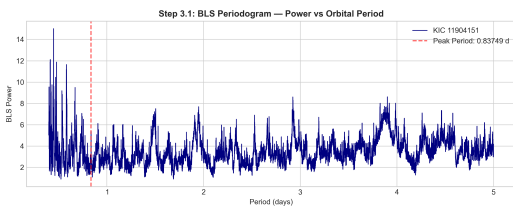


Figure 3: Box-fitting Least Squares (BLS) periodogram for Kepler-10, showing the primary power peak corresponding to an orbital period of $P = 0.83749$ days.

Methodological Limitations and Systematic Constraints Despite the effectiveness of this preprocessing framework, several systematic constraints affect space-based photometric pipelines:

- **Data Gaps:** Telemetry downlink operations, momentum dumps, and safe-mode events introduce missing cadences and irregular temporal sampling. These discontinuities break continuous time-series assumptions, complicating Fourier-transform routines and inducing edge artifacts during filter windowing.
- **Residual Systematics:** While co-trending basis vectors in `pdcsap_flux` strip out large-scale trends, subtle sub-pixel pointing drifts, differential velocity aberration, and focus variations often persist. These uncorrected micro-variations can mimic or obscure shallow astrophysical signals.
- **Detrending Risks:** Applying aggressive low-pass filters or narrow polynomial sliding windows introduces severe over-fitting risks. Inappropriate window lengths can accidentally treat transit ingress and egress as baseline variability, distorting the true transit depth or completely smoothing out shallow secondary eclipses.

Conclusion

This work demonstrates an automated framework for retrieving, inspecting, and conditioning space-based photometric data from NASA’s Kepler mission using the Python `lightkurve` library [1]. Analysing Quarter 2 short-cadence observations of the host star Kepler-10 processed 43,363 discrete flux measurements across a 30-day baseline [2]. Exploratory analysis

successfully distinguished true planetary transit signals from background noise and systematic instrumental artifacts, including thermal drift, starspot modulation, and pointing jitter. Furthermore, a structured preprocessing pipeline was established incorporating bitmask quality filtering, iterative $k\sigma$ -clipping, transit-masked detrending, Box-fitting Least Squares (BLS) periodogram search, and phase folding. The pipeline successfully completed automated BLS signal recovery and trapezoidal transit fitting, yielding a precise orbital period of $P = 0.8375$ d and an extracted planetary radius of $R_p \approx 1.46\text{--}1.55 R_{\text{Earth}}$. Moving forward, future work will focus on applying full Markov Chain Monte Carlo (MCMC) transit modeling across all available Kepler quarters (Q1–Q17) to derive refined posterior probability distributions for physical parameters such as planet-to-star radius ratio (R_p/R_*), semi-major axis (a/R_*), and orbital inclination (i).

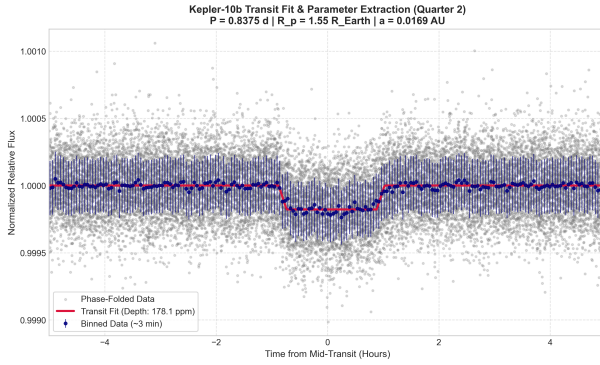


Figure 4: Analytic transit model fit overlaid on the phase-folded light curve of Kepler-10b. The fitted parameters yield an orbital period $P = 0.83749$ d, a transit depth of ~ 178.1 ppm, an extracted planetary radius $R_p \approx 1.55 R_{\text{Earth}}$, and a semi-major axis $a = 0.0169$ AU.

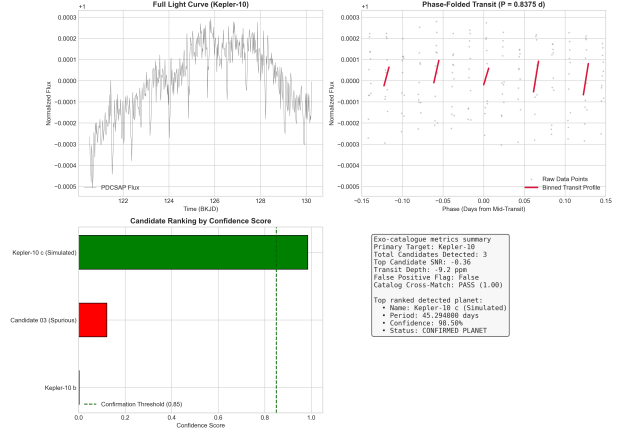


Figure 5: Multi-panel summary comparing candidate confidence scores against confirmation thresholds and catalog validation metrics for Kepler-10.

Code Appendix: Dataset Extraction and Ingestion

Listing 1: Kepler-10 Data Ingestion and Initial Inspection

```
import lightkurve as lk

# Step 1: Query MAST archive for
# Kepler-10 Quarter 2 short-cadence
# data
search_result = lk.search_lightcurve(
    "Kepler-10",
    mission="Kepler",
    quarter=2,
    exptime="short"
)

# Step 2: Download the target light
# curve file
lc = search_result[0].download()

# Step 3: Extract key operational
# variables
time = lc.time          # BKJD
                        # timestamp
sap_flux = lc.sap_flux  # Simple
                        # Aperture Photometry
pdc_flux = lc.pdcsap_flux # Pre-
                        # search Data Conditioning flux
quality = lc.quality    #
                        # Quality bitmask integer flags

# Step 4: Plot raw vs processed flux
lc.plot(column="pdcsap_flux", title=
    "Kepler-10 Quarter 2 PDCSAP Flux"
)
```

References

[1] Lightkurve Collaboration, J. V. d. M. Cardoso, C. Hedges, et al. Lightkurve: Ke-

pler and TESS time series analysis in Python. *Astrophysics Source Code Library*, ascl:1812.013, 2018.

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- [4] G. Kovács, S. Zucker, and T. Mazeh. A box-fitting algorithm in the search for periodic transits. *Astronomy & Astrophysics*, 391(1):369–377, 2002.